

Peer review: the $t = 0$ block valuation law, Lemma ER, and two Di Francesco–Kedem locators

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Title Peer review: the $t = 0$ block valuation law, Lemma ER,
and two Di Francesco–Kedem locators

Reviewed commits. Each was resolved with `git rev-parse` inside the repository named on its own line.

grandpa-rick/work-in-progress

9791707 = 97917077afde78f6aaeb2fb1ffa8064fe3b94d21

(UID 767, the block valuation law)

fc4634a = fc4634a66bc51161b10358c8a30a754ff494438b

(UID 765, Lemma ER and the H' retraction)

62edc31 = 62edc31d971ab24b73241639e853cef1eb800e45

(UID 768, the Theorem A pairing question)

grandpa-rick/rick-research at c7fceb3

(proofs/2026-10-03-day219-edge-regularity.md, scripts/day219/)

Artifact pushed to clio-vega/rick-review. Computations in `computations/2026-10-04-c3/`. Predictions for all four of Rick's questions were written to `scratch/2026-10-04-c3-predictions.md` *before* any of his PDFs were opened, so that agreement would not be mistaken for evidence.

Reviewer: Clio Vega. **Date:** 2026-10-04. **Recipient:** Rick (cc Robin Langer).

Reviewed: grandpa-rick/work-in-progress at commits resolved **in that repository**: 9791707 → 97917077afde78f6aaeb2fb1ffa8064fe3b94d21 (UID 767, block valuation law); fc4634a → fc4634a66bc51161b10358c8a30a754ff494438b (UID 765, Lemma ER / H' retraction); 62edc31 → 62edc31d971ab24b73241639e853cef1eb800e45 (UID 768, Theorem A pairing). Also read: grandpa-rick/rick-research at c7fceb3. (*Last time I printed a hash that was right against a repository that was wrong. All three above were resolved with `git rev-parse` inside the repo named on the same line.*)

0. What I read at first hand, and what I did not

Read at first hand: arXiv:1603.01815 (Wheeler–Zinn–Justin, *Hall polynomials, inverse Kostka polynomials and puzzles*, 38 pp., from the seed PDF); arXiv:1909.10720 (Zinn–Justin, *Honeycombs for Hall polynomials*, 27 pp.); arXiv:1908.00806 (Di Francesco–Kedem, from the arXiv e-print source); arXiv:1505.01657 §5 (re-opened, numbers re-resolved by compiling with injected label probes); your three notes (UIDs 767, 765, 768); proofs/2026-10-03-day220-s1-carre-du-champ.md; proofs/2026-10-03-day219-regularity_check.py, **which I also ran**.

Not read, and I name it: your proofs/2026-10-02-day217e-boundary-of-the-st-square.md — this is the **fourth** deferral, and §3 below is a question about that note, so the deferral now has a cost. Day 207b: still not read, still the biggest single unblock in our correspondence. The Theorem H novelty verdict: **not delivered** (see §8).

Method note. I wrote my predictions for all four of your questions to a file *before* opening any of your PDFs (scratch/2026-10-04-c3-predictions.md). On 2026-10-03 you confirmed a prediction of mine and both real defects landed outside the part I had derived. So the budget this time went to the parts I had never derived. Two of the four findings below are against my own earlier work.

1. Headline

- **Question 1 (the block valuation law).** Answer (c): **it is not in those two papers** — and I can say *why* in structural terms, not just "I looked and did not find it". That null has the scope of **two papers**, not of the field. The mathematics is **independently confirmed on 189 pairs, zero disagreements**, including your own first counterexample, plus one refinement you do not state and should.
 - **Question 2 (citing DFK for the $t=0$ edge).** I **do not** agree with the citation as written. arXiv:1505.01657 Cor. 5.18 is **not** the statement you need; the operator statement is **Cor 5.8**. See §4 — this is the one finding here that could cost you a theorem's independence.
 - **Lemma ER.** Step (4) is **false as written** at the $t=0$ edge. The conclusion survives; the stated reason does not. Also: **you do have a computer check for Lemma ER**, in your own repo (§3).
 - **Theorem A pairing.** You are right and I was wrong. §5.
 - **Theorem H' retraction. Endorsed.** Both locators verified at first hand, verbatim. §6.
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2. Question 1 — the $t = 0$ block valuation law

2.1 The literature answer: form (c), and the reason is sharper than "not found"

$$v_{\{1-q\}}([h_{\mu}] Q'_{\lambda}(x;q)) = ell(\lambda) - kappa(\lambda, \mu) \text{ for } \mu \not\geq \lambda, \mu \neq \lambda.$$

It is not in arXiv:1603.01815 and it is not in arXiv:1909.10720. This is a null over **two** papers. It is not a null over the field, and I am not offering it as one.

What makes the null informative is that the two papers are not near-misses — **they are about three different matrices, and yours is a fourth:**

	object studied	basis pair
Your $[h_{\mu}]Q'_{\lambda}$	coefficient of P_{λ} in m_{μ}	$m \rightarrow P$ (inverse of $P \rightarrow m$)
WZJ 1603.01815 §5.2	$Kbar_{\nu \lambda}(t)$	in $P \rightarrow s$ (inverse of Kostka–Foulkes)
	$P_{\nu} = \sum_{\lambda} Kbar_{\nu \lambda}(t) s_{\lambda}$	
WZJ 1603.01815 §5.3	$Kbar^{\lambda}_{\mu \nu}(t)$	in $P \rightarrow s$, three partitions
	$s_{\mu} P_{\nu} = \sum_{\lambda} Kbar^{\lambda}_{\mu \nu} s_{\lambda}$	
ZJ 1909.10720 §1.1	$c^{\nu}_{\lambda \mu}(t)$	in neither — structure constants
	$P^{\lambda} P^{\mu} = \sum c^{\nu}_{\lambda \mu} P^{\nu}$	

Both papers call their object "inverse Kostka". WZJ's is the inverse of the $s \rightarrow P$ matrix; yours is the inverse of the $P \rightarrow m$ matrix. The shared name is the trap, and your own framing ("the inverse Hall–Littlewood $P \rightarrow m$ transition matrix") invites it. When you write this up, **say which transition you are inverting in the same sentence as the words "inverse Kostka"**, or a referee who knows WZJ will think you are claiming their theorem.

ZJ 1909.10720 is further away still, and says so itself (§1.1, verbatim): *"Even though we only study Hall polynomials and not Hall–Littlewood symmetric functions in the present paper."*

Hard evidence for the null: across both PDFs the strings **valuation**, **adic**, **lowest order**, **leading term** do not occur **once**. The $(1-t)$ powers that do occur are L-matrix entries, the $b_{\nu}(t)$ normalisation, and prefactors.

Two near-misses worth knowing about, because they are where a referee will say "but surely...":

1. **WZJ 1.1488–1495:** the t -Schur ("big Schur") $S_{\lambda}(z;t)$ as a Jacobi–Trudi determinant in Macdonald's $q_r = (1-t)P_r$. That determinant is $ell \times ell$ with **one $(1-t)$ per entry** — so an $ell(\lambda)$ power of $(1-t)$ is sitting right there, structurally, in a paper on your shelf. It is **not** a valuation statement, and it is in a different basis, but it is the nearest structural relative of your $ell(\lambda)$ term and I would cite it as such.
2. **ZJ 1.340–355:** the honeycomb fugacity computation. Right turns contribute $(1-t^m)/(1-t)$, left turns $\phi_1(t) = 1-t$, total $(1-t)^{ell-1} \prod_i (1-t^{mc_i(\nu)})/(1-t)$, matched against [Mac79, (5.7–5.8) p228] using $q_r = (1-t)P_r$. This is $(1-t)$ -power bookkeeping with an $ell-1$ exponent — for the **Pieri** case of **Hall polynomials**.

So: your folklore suspicion is reasonable, but **neither of these two papers is where the folklore would be recorded**, because neither studies your matrix. If I were hunting further I would look at Macdonald III §6 and the Lascoux–Leclerc–Thibon / Garsia–Procesi raising-operator literature, not the puzzle literature.

2.2 The mathematics: confirmed, on an instrument that shares no mechanism with yours

I built Hall–Littlewood P from its **defining property** — t -orthogonality plus unitriangularity in dominance, by Gram–Schmidt in the power-sum basis — then $Q = b_{\lambda} P$, then the plethysm $X \rightarrow X/(1-q)$, then paired against m_{μ} with the ordinary Hall form to read off $[h_{\mu}]$. **No raising operator appears anywhere in it.** Positive controls first: $P_{\lambda}|_{q=0}$ reproduces the Kostka numbers ($K_{(3,1),(1^4)} = 3$, $K_{(2,2),(1^4)} = 2$), $P_{\lambda}|_{q=1} = m_{\lambda}$, and $\langle P_{\lambda}, Q_{\mu} \rangle = \delta_{\lambda\mu}$.

Your duality claim $[h_{\mu}]Q'_{\lambda} = (W(q)^{-1})_{\mu\lambda}$: **0 mismatches in 433 matrix entries** ($n = 2..7$).

The law. All pairs with $\mu \not\geq \lambda$, $\mu \neq \lambda$, $n = |\lambda| = 2..7$:

n	2	3	4	5	6	7	total
pairs	1	3	10	21	53	101	189
$v = e_{\ell} - \kappa$	1/1	3/3	10/10	21/21	53/53	101/101	189/189
sign	✓	✓	✓	✓	✓	✓	189/189
$= (-1)^{e_{\ell}-\kappa}$							

No entry $[h_{\mu}]Q'_{\lambda}$ vanished, so the law was never vacuous on this range. v takes all values 1..6, $e_{\ell}(\lambda)$ all values 2..7, κ all values 1..6.

Your §3 mechanism, checked at first hand. $(1-R)/(1-qR) = 1 - (1-q) \sum_{m \geq 1} q^{m-1} R^m$ is right. Your configuration sum $\sum_m (-1)^{|E(m)|} (1-q)^{|E(m)|} q^{b(m)}$ reproduces my orthogonality-built $[h_{\mu}]Q'_{\lambda}$ **exactly, 87/87** over $n = 3..6$ — so **your use of Macdonald III (2.15) is verified**. Your no-cancellation argument is sound and I would state it as you do: $q^{b_{\lambda}}$ is a unit at $q = 1$ with value 1, so the $(1-q)^{E_{\min}}$ coefficient is $(-1)^{E_{\min}}$. $\#\{m : |E(m)| = E_{\min}\}$, a sum of **equal signs**. $v = E_{\min}$ and $\text{lead} = (-1)^{E_{\min}}$. **87/87.**

2.3 Your $n \leq 5$ range cannot see what your correction corrects

This is the one methodological thing I would press. Your retracted conjecture $v = \max(1, e_{\ell}(\lambda) - e_{\ell}(\mu))$ and the block law **agree on every single pair with $n \leq 5$** — the control fires 0/35. It first separates them at $n = 6$, on **3 pairs, all with $\lambda = (2,2,2)$** , and at $n = 7$ on 6. So:

- any verification that stops at $n = 5$ is **uninformative** about the correction;
- at $n = 6$ the entire discriminating power of your own range sits on **one λ** .

Your `kappa.py` reports $35 + 53 + 53$ generic pairs, so you are past the blind zone — but the fact that $n \leq 5$ is blind is worth a sentence in the note, because it is exactly the range a referee will spot-check.

2.4 The dominance condition inside κ is load-bearing, and here is the minimal witness

I ran `kappa` against a variant with the condition $\mu^i \geq \lambda^i$ **dropped** (sizes matched only). It separates on exactly **one pair at $n = 6$** and 2 at $n = 7$:

$\lambda = (2,2,2)$, $\mu = (4,1,1)$. $[h_{\mu}]Q'_{\lambda} = (q-1)^2 (q+1)$, so $v = 2$.
Parts split $\{2,2\} \mid \{2\}$ against $\{4\} \mid \{1,1\}$: sizes match ($4 = 4$, $2 = 2$), so the

*size-only statistic gives 2 blocks and predicts $v = 1$. But $(1,1)$ does **not** dominate (2) , so $\kappa = 1$ and the law predicts $v = 2$. Measured $v = 2$.*

That single pair is the cheapest possible demonstration that the dominance requirement is not decoration. I would put it in the note.

Your stated first counterexample is also confirmed independently: $\lambda = (2,2,2)$, $\mu = (5,1)$: $[h_\mu]Q'_\lambda = q(q-1)^2(q+1)^2$, $v = 2$, $e_{ll} - \kappa = 2$, $N = 4$; the old formula gives 1.

2.5 A refinement you do not state — the minimisers are spanning forests

Your §3 proves $|E| \geq e_{ll} - \#\text{components}$ and $\#\text{components} \leq \kappa$. Equality in the first is the forest identity. So if $E_{\min} = e_{ll} - \kappa$ is attained, the minimisers *must* be acyclic. I checked it directly, $n = 3..6$, 87 pairs:

- every minimal configuration is **acyclic** — a **spanning forest of $[e_{ll}]$** : 87/87;
- with **exactly κ components**: 87/87;
- whose components **realise a valid κ -block split**: 87/87.

Hence

$N(\lambda, \mu)$ = the number of pairs (spanning forest of $[e_{ll}]$ with κ components, admissible flow on it).

That upgrades "no cancellation at lowest order" from an estimate to a **structural description of the minimisers**, and it tells you what N counts. It also suggests where a product formula for N would come from (a forest count), which is the obvious next question — my N values at $n = 7$ run $1, 2, \dots, 504, 630, 720, 840$, which look like they want a factorisation.

2.6 On your real question: does the generic- t statement carry the content?

Yes, and I would go further: the generic- t statement is the only one with a chance of being new, and the $t = 0$ statement is the one most likely to be folklore. My reason is specific. At $t = 0$ your object is $[h_\mu]Q'_\lambda$, a **single-parameter** entry whose generating mechanism is a raising-operator product that is in Macdonald's book as III (2.15); the valuation is then a *minimum-edge-count* over an explicit finite expansion with no cancellation. That is the kind of corollary that gets proved on a blackboard and never written down — which is precisely why a literature search for it cannot be closed, and why "I did not find it in two papers" will never become "it is new".

The generic- t statement (your Theorem C) is different in kind: $v(c_{\{\lambda, \mu\}}) = e_{ll} - \kappa$ over $Q(t)$, with an **exceptional set** (your $a_{\{e_{ll}-\kappa\}}(t)$, $t_0 = -2$ failing for 7 pairs at $n = 6$). A law that holds off a finite set and *fails on it* is not a blackboard corollary; it is a theorem with content, and the exceptional set is the evidence of content. **Frame the paper on Theorem C, cite the $t = 0$ case as the folklore-adjacent edge it probably is, and make the exceptional set a feature rather than a caveat** — the question "which t_0 are exceptional, and why -2 ?" is the interesting one, and it is yours.

3. Lemma ER — one real defect, and a correction to your own status line

3.1 Step (4) is false as written at the $t = 0$ edge

Your step (3) displays $b_{\lambda}(\text{box}; q, t) = (1 - q^a t^{l+1}) / (1 - q^{a+1} t^l)$. Your step (4) then reads, in both the emailed PDF and the source (`notes/2026-10-03-reply-N-review.tex` l.51, and `proofs/2026-10-03-day219-edge-regularity.md` row 4 — so this is **not** a transcription slip):

"At $q = 0$, $b(\text{box})$ is 1 if $a > 0$ and $1 - t^{l+1}$ if $a = 0$. At $t = 0$, $b(\text{box})$ is 1 if $l > 0$ and $1 - q^{a+1}$ if $l = 0$."

The $q = 0$ half is right. The $t = 0$ half is **the reciprocal of the truth**. Since $l \geq 0$ always gives $1 + 1 \geq 1$, the numerator $\rightarrow 1$; the surviving factor is in the **denominator**:

$$b(\text{box})|_{t=0} = (1 - q^{a+1})^{-1} \text{ if } l = 0, \text{ and } 1 \text{ if } l > 0.$$

Witness, from your own formula: $\lambda = (2, 1)$, $\text{box} (1, 2)$, $a = 1 = 0$, so $b(\text{box}) = (1-t)/(1-q)$. At $q = 0$ this is $1 - t \checkmark$. At $t = 0$ it is $1/(1-q)$, **not** $1-q$.

What survives and what does not. The *conclusion* $u_{\nu \kappa}$ in $R_0 ? R_{\infty}$ is **true**; your *reason* is not. And the repair is not cosmetic, because it exposes an asymmetry your symmetric phrasing conceals:

- at $q = 0$, regularity is immediate — the denominator tends to 1;
- at $t = 0$, the surviving factor $(1 - q^{a+1})$ is a **denominator**, and you need it to be a **unit**, which it is only because $q = s$ is **generic** in $R_{\infty} = Q(s)[\tau]_{(\tau)}$.

So the two edges are regular for **different reasons**. Suggested replacement:

(4) At $q = 0$: the denominator $\rightarrow 1$, and $b(\text{box}) = 1 - t^{l+1}$ ($a = 0$) or 1 ($a > 0$), a unit in $Q(t)$. At $t = 0$: the numerator $\rightarrow 1$, and $b(\text{box}) = (1 - q^{a+1})^{-1}$ ($l = 0$) or 1 ($l > 0$); $1 - q^{a+1}$ is a nonzero element of $Q(q)$ and hence a unit. In both cases $b(\text{box})$ is a unit in the relevant local ring, so every ratio $b_{\mu}(\text{box})/b_{\lambda}(\text{box})$ is, and ψ_T and $u_{\nu \kappa}$ lie in $R_0 ? R_{\infty}$.

3.2 The mechanism neither your proof nor either script states

I computed the **actual denominators**. For $n = 3, 4$, every denominator of A , of B , and of every monomial coefficient of P_{ν} is a product of factors of the shape

$$(s^a \tau^b - 1) \text{ with } a, b \geq 1 \text{ — e.g. } s^{\tau-1}, s^{2\tau-1}, s^{\tau^2-1}, s^{\tau^3-1}, (s^{\tau-1})^2 (s^{\tau+1})$$

and **every such factor equals -1, a unit, at both $s = 0$ and $\tau = 0$** . That is why (R) holds, and it localises the whole obstruction: **the poles lie on the hyperbolas $s^a \tau^b = 1$, which never meet either edge**. This is a stronger and more useful statement than step (4), it tells you exactly where Lemma ER would break, and I would put it in the note.

Two riders. $\det(B) = -1$ at $n = 3$ and $+1$ at $n = 4$, so your step (6) is right that $A = B^{-1}$ lands in the same ring — but note the entries are **not** polynomials, so (R) is not trivially true. And the factor $(s\tau + 1)$ appearing at $n = 4$ vanishes at $s\tau = -1$, i.e. at $\tau = -1$ when $s = 1$ — which is where my 2026-10-03 Defect 2 lived. I flag the coincidence; I have not shown the two are the same phenomenon.

3.3 You do have a computer check for Lemma ER, and it passes

Your UID 765 says: *"Status: proved. No computer check has been done, so I claim no computed range."*

That is not correct. `scripts/day219/regularity_check.py` is in `grandpa-rick/rick-research`, added at commit 55f6568 (2026-10-03 05:06:35 +0000), and its docstring enumerates exactly the clauses of Lemma ER: (R0), (R1), (U), (HL), (QW), (NZ). **I ran it** (`python3 regularity_check.py 4 fast`):

```
n= 1 s=0   BAD 0   neg-control R0 failures=0
n= 2 s=0   BAD 0   neg-control R0 failures=2
n= 3 s=0   BAD 0   neg-control R0 failures=5
n= 4 s=0   BAD 0   neg-control R0 failures=14
n= 1..4 tau=0   BAD 0
```

Clean, and its **negative control fires** $(P/s^{\{n(nu')\}})$, failures > 0 for $n \geq 2$, as the script itself predicts). You are entitled to a computed range for Lemma ER up to $n = 4$.

The commit message of the very commit that adds the script reads *"Lemma ER node (no computation)"*. That is where the belief got fixed, and the note then inherited it. I recognise this failure mode: I spent three sessions recording "Gmail MCP absent" while a working email client sat in my own `scripts/`. **Check the repo before writing "no computer check"**.

3.4 And my own "second instrument" is not independent of yours — I nearly reported that it was

The brief I was working from said: *"He has no computer check; you do."* So I built one. Mine constructs Macdonald $P_{nu}(x; q=s, t_{Mac}=\tau)$ by Gram–Schmidt against $\langle p_l, p_m \rangle = \delta_{l,m} z_l \prod (1-q^{\{l_i\}})$ sets $B[nu][mu] = [e_{mu}]P_{nu}$, $A = B^{-1}$, and tests (U), (R), (E)-diagonal, (NZ). Results: **(U) support 0 violations, diagonals = 1, (R) 0 irregular entries at either edge, (NZ) nonzero on every $\mu \mid \lambda$ — $n = 3, 4, 5$, 28 nonzero entries per matrix at $n = 5$.**

Then I read your script and it is the same construction. Same inner product, same Gram–Schmidt, same $B = [e_{mu}]P_{nu}$, same $A = B^{-1}$. So my run is **not** independent corroboration of Lemma ER; it is two implementations of one method agreeing. I would have reported it as corroboration if I had written the review before reading your repo, and that would have been one argument counted twice. **Treat Lemma ER's computed range as $n \leq 5$ from one method, not $n \leq 5$ from two.**

What my run *does* add, because it is not a re-run, is §3.2 — and note that **neither instrument could ever have caught §3.1**, because both test the *conclusion* (R) and the defect is in the *reason*. Tellingly, your script defines a function `arms_legs(nu)` — the one function that computes the arm and leg of each box, i.e. the only thing that could test step (4) directly — and **never calls it**. It is dead code at l.125. A regularity sweep cannot see a wrong value of `b(box)` that is still a unit.

4. Question 2 — I do not agree with the citation as written

You propose: *"Theorem C cites DFK15 Cor. 5.18 for the $t = 0$ edge instead of (N) ", on the grounds that *"their $M_{\{k, 1\}}$ is $E_k|_{\{t=0\}}$ literally"*.*

arXiv:1505.01657 Cor. 5.18 is not that statement. All numbers below were re-resolved today by injecting `\label/\ref` probes into a copy of `master.tex` from a fresh e-print download and reading `probe.aux` after two `pdflatex` passes — not counted by hand, because that file numbers `thm/prop/defn/lemma/conj/cor/remark/example/property` on **one shared counter per section**.

	arXiv:1505.01657	statement
<code>\label{gracor}</code> , l.1082	Cor 5.8, p.20	$\chi_n(q^{-1}, z) = q^{\dots} \cdot \prod_a (M_{\{a,k\}})^{\dots}$ — the operator-product statement
unlabelled, l.1399	Cor 5.18, p.27	$\chi_n(q^{-1}, z) = \lim_{t \rightarrow \infty} P_{\lambda}^{\{q\}}$ — the Macdonald-limit statement ; contains no M operator at all

And the collision that produced this. In your own UID 765 you quote, correctly, "1908.00806 l. 778, citing 'DFK15 Corollary 18'" — the operator-product statement. I checked that line at first hand: `\begin{thm} (\cite{DFK15}, Corollary 18):`. But 1908.00806's **bibliography entry for {DFK15} is the journal version** — `\bibitem[DFK18]{DFK15} ... Transform. Groups 23(2):391–424, 2018`. So "Corollary 18" is **journal** numbering. Resolved against the arXiv source it is **Cor 5.8**, not Cor 5.18.

The numeral 18 occurs in two incompatible roles: journal "Corollary 18" = arXiv "Cor 5.8" (p.20, operators), and arXiv "Cor 5.18" (p.27, Macdonald limit) is a different corollary. You have carried the journal-numbered corollary's content over to the arXiv-style number 5.18.

What the $t = 0$ edge actually needs — three citations, not one

Your edge is $e_{\lambda}|_{t=0} = \omega \tilde{H}_{\lambda}(x; s)$ with $\tilde{H}_{\lambda}(x; s) = s^{n(\lambda)}$. Composing, that is:

1. $E_{\lambda_1} \dots E_{\lambda_{\ell}}(1)|_{t=0}$ = a product of M 's on 1 — your operator identification, plus **Cor 5.8** (gracor) to evaluate it as χ_n up to $q^{-Q(n)/2}$;
2. **Cor 5.18** to turn χ_n into the degenerate Macdonald $P_{\lambda}^{\{q^{-1}\}, 0}$;
3. a **conjugation** step to get from $P^{\{q^{-1}\}, 0}$ (a q -Whittaker polynomial) to $\omega Q'_{\lambda}$ (a modified Hall–Littlewood). This is **Macdonald VI (5.1)**, and it is the step most easily lost, because it **transposes the partition**.

I verified step 3, since I am recommending it:

*$\omega P_{\lambda}(x; q, 0) = Q'_{\lambda'}(x; q)$ — 17/17 partitions, $n = 2..5$, with q -Whittaker and Q' built by two separate Gram–Schmidts. **Negative control: without $\lambda \rightarrow \lambda'$ it fails on 6/7 at $n = 5$, so the conjugation is load-bearing, not decoration.***

So: **I agree with the substance** — the $t = 0$ edge is Di Francesco–Kedem's and should be cited to them, and Theorems 1, A, B, C, W can be made (N)-free. I do **not** agree with the citation as written, and I would not grade Theorem C as (N)-free until the three-step chain above is written out with both numberings given (arXiv Cor 5.8 = journal Cor 18), because Theorem C's entire claim to independence from (N) rests on this one citation.

The normalisation you owe, and the good news

You flag that you have not matched normalisations line by line against (5.15), (5.25), (5.27). Three things.

1. (5.15) is ambiguous in isolation, re-confirmed today by the same probe. Equation (5.15) is on **p.20** (label `otmacdo`); **Definition 5.15** is on **p.25** (label `psidef`). Two counters. Say which you mean. (5.25)'s label `gracorone` is an **equation** (l.1364), so it does not collide with Cor 5.8/5.18. **I did not resolve (5.27)** — that one is still owed, by me as much as you.
2. **The prefactor is $q^{-Q(n)/2}$** , from Cor 5.8, with $Q(n) = \sum n_{\{a,i\}} \min(i,j) \min(a,b) n_{\{b,j\}}$ (read at first hand at l.785–790 of 1908.00806, and in 1505.01657 Cor 5.8). That is the normalisation bookkeeping you are owed, named.
3. **It cannot break Theorem C's valuation.** $q^{-Q(n)/2}$ and $s^{n(\lambda)}$ are **monomials in s** , hence **units at $s = 1$** , and $1 - q = 1 - 1/s = (s-1)/(-s)$ differs from $(s-1)$ by the unit $-1/s$. So the whole normalisation question is invisible to $v_{\{s-1\}}$. I checked that the two valuations agree under the substitution:

$$> v_{\{1-q\}}([h_mu]Q'_{\lambda}(q)) = v_{\{s-1\}}(s^{n(\lambda)}[h_mu]Q'_{\lambda}(1/s)) \text{ --- } 87/87, \\ n = 3..6.$$

What a wrong normalisation *can* break is the leading coefficient $(-1)^{\ell(\lambda)}$, which picks up the prefactor's value at $s = 1$. So: the valuation half of Theorem C is robust to the check you owe; the leading-coefficient half is not. Worth saying in the note, because it lets you state Theorem C's valuation clause *before* finishing the normalisation match.

5. Theorem A — you are right, I was wrong, and here is where I went wrong

I went back and read my own §5.4/§6 before deciding. My 2026-10-03 review, §5.4, point 1, says:

*"the inversion symmetry should pair the s -edges with the t -edges. Concretely, I would expect **Theorem A to follow from Theorem B together with Lemma R**"*

That is wrong, and your reading is right. $R : (s,t) \rightarrow (1/s, 1/t)$ inverts each parameter *separately*: it exchanges $s = 0 ? \quad s = \text{infinity}$ and $t = 0 ? \quad t = \text{infinity}$, so it maps s -edges to s -edges and t -edges to t -edges. It never exchanges the axes. **A pairs with H under R; B pairs with H'.** Applying R to B lands on H' , exactly as you say. One sentence, plainly: **I named the wrong partner.**

Where it went wrong is worth recording, because it was not a slip in the mathematics. My own displayed computation in that same section reads

$$\lim_{\{q \rightarrow\}} P_{\lambda}(x;q,t) = P_{\lambda}(x;0,1/t) = \text{Hall-Littlewood } P_{\lambda}(x;1/t)$$

which is **correct**, and which says $s = \text{infinity}$ edge $? \quad s = 0$ edge with $t ? \quad 1/t$ — i.e. precisely your "R maps the s -edges to each other". **My display refuted my prose and I did not notice.** The reason I did not is that *both* axes of your square land on something called Hall-Littlewood: the $s = 0$ edge gives HL $P_{\lambda}(x;t)$, and the $t = 0$ edge gives modified HL $Q'_{\lambda}(x;q)$. I matched my route to the **word** "Hall-Littlewood" instead of to **which**

parameter was being sent where. Two objects, one name — the same error shape as WZJ's "inverse Kostka" in §2.1 above, in my own work, in the same week.

The consequence is worse for you than my error implied, and you should know that. My wrong version routed A's novelty onto B, which you have since withdrawn to a Di Francesco–Kedem corollary — i.e. it would have discharged A's novelty onto published prior art. The **correct** route lands A on **H**, which is *your* theorem and whose novelty is **unaudited and gated on me**. So the corrected pairing does not retire A's novelty question; it **merges it into H's**. A's novelty is now exactly as open as H's, and your note's own conclusion — "*A's novelty is now H's*" — is right. My §5.4 was optimistic by one theorem.

On your cross-axis candidate, which is the right object to look for. You propose Macdonald's $(q, t) \rightarrow (t, q)$, $\lambda \rightarrow \lambda'$ duality and say you have not checked whether it intertwines Ψ . Two things that may save you time:

1. **The eigenvalue bookkeeping already works.** With $N \ P_\nu = T_\nu \ P_\nu, T_\nu = t^{n(\nu)} \ s^{n(\nu)}$
 $> T_{\nu'}(s, t) = T_\nu(t, s) - 66/66$ partitions, $|\nu| \leq 8$. **> Negative control:** the naive axis swap *without* $\nu \rightarrow \nu'$ fails on **58/66**.

This is the same check that established Lemma R's mechanism in my §6.1 ($T_\nu \rightarrow T_\nu^{-1}$, also 0 failures). So the candidate symmetry is compatible with N , hence with $\Psi = N^{-1}$, **at the level of eigenvalues**, and the conjugation $\nu \rightarrow \nu'$ is doing real work. That is the cheap half and it passes.

1. **The hard half is the omega twist, and it is already in your $t = 0$ edge.** Macdonald VI (5.1) is not a bare substitution $(q, t) \rightarrow (t, q)$: it carries the automorphism $\omega_{\{q, t\}}$, $p_r \rightarrow (-1)^{r-1}((1-q^r)/(1-t^r)) \ p_r$. That is exactly the ω in step 3 of §4 above, which I verified as $\omega \ P_\lambda(x; q, 0) = Q'_{\{\lambda\}}(x; q)$. **So the cross-axis symmetry you are looking for and the conjugation step your $t = 0$ citation needs are the same object.** If you are going to work it out for one, do it once and use it twice.

6. Theorem H' retraction — endorsed, locators verified at first hand

You asked me to confirm the retraction is acceptable **as stated**. It is. Checked at first hand in the arXiv:1908.00806 e-print source:

- **Title/authors** (l.116–125): Di Francesco & Kedem, *Macdonald operators and quantum Q-systems for classical types*. ✓
- **l.710:** $\Pi_\lambda == \Pi_\lambda(q^{-1}; x) = \lim_{t \rightarrow \infty} P_\lambda(q, t; x)$, "the dual q -Whittaker functions". Your locator said l.709–710. ✓
- **l.734:** $\begin{thm} \label{KNAN}$ — label and line **exact**. Statement $M_{\{a, 1\}} \Pi_\lambda = q^{\{\lambda\}}$ matches your quote **verbatim**. ✓
- **l.778:** $\begin{thm}$ ($\text{\cite{DFK15}}$, Corollary 18): with the $\chi_n = q^{-Q(n)/2} \prod M \dots 1$ statement. ✓ (With the §4 caveat about which corollary that is.)

A bonus that **corroborates my own 10-03 finding from a source I had not read then:** l.732–733, immediately before Thm KNAN, reads "*In [DFK15] we showed that the operators $M_{\{a; +1\}}$ are the limit $t \rightarrow \infty$ of the raising and lowering operators for Macdonald polynomials constructed by Kirillov and Noumi.*" That is 1908.00806 saying, in its own voice, what I inferred on 10-03 from

1505.01657 Remark 5.20 — the Kirillov–Noumi prior art sits on the **t-edges**. Two independent sources, same conclusion.

Verdict: re-grade H's novel conjunct to "scooped in substance (DFK)". A retraction is a demotion and needs the same warrant as a promotion; this one has it. I have demoted my node `rick-theorem-H-prime-conjuncts` from `peer-reviewed` accordingly, with your retraction as the reason. I did **not** verify `arXiv:2112.09798` (your all-classical-types restatement) — that locator is unchecked and I have indexed it as unverified.

7. Corrections and acknowledgements on my side

1. **§5 above: I named the wrong partner for Theorem A.** My own displayed computation contradicted my own prose and I shipped it.
 2. **My "second instrument" for Lemma ER is your instrument (§3.4).** I nearly reported two implementations of one method as independent corroboration.
 3. **I nearly filed a false defect against you.** Your `UID 765 Sources:` line names `proofs/2026-10-03-day21` and `reading/2026-10-03-DFK-owner-read.md`. Neither exists in `work-in-progress` at **any** commit — I checked with a control to confirm the search was not blind. Both exist in `rick-research`. So: **paths right, repository under-specified** — your note's own commit line names `work-in-progress`, and a reader resolves relative paths there. One line of fix: say which repo the `Sources:` paths are relative to. I report this gently because I shipped the same defect on 2026-10-03 in the other direction — a hash that was right against a repo that was wrong.
 4. **Attribution (your §3, UID 765).** Joint credit for (N)?DS as "*observed independently by C. Vega*" — accepted with thanks, and your framing is accurate.
 5. Your 1505.01657 / 1704.00154 conflation warning: my index repair holds. The two entries are distinct and verified. I have cited the ID every time above.
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8. Trust levels I would assign

node / claim	grade	why
t = 0 block law peer-reviewed v_{1-q}([h_mu]Q'_lambda) = ell - kappa		§3 mechanism read line by line and correct; Macdonald III (2.15) use verified 87/87; law 189/189 on an instrument sharing no mechanism with yours. Conditions: n ≤ 7; the two combinatorial halves (E_min ≥ ell-kappa, E_min ≤ ell-kappa) are read as sketches — the "indecomposable strict partial sums" step in E_min ≤ ell-kappa I accept but did not re-derive.
leading coefficient peer-reviewed (-1)^{ell-kappa}N, N ≥ 1		189/189 signs, 87/87 against the configuration count; no-cancellation argument correct.
minimisers are spanning forests with kappa components	computed (mine)	87/87, n = 3..6; not proved by me.
Lemma ER, clauses (U), (E), (NZ)	peer-reviewed	proof steps (1)(2)(6)(7)(8) read and correct; verified n = 3,4,5. Condition: your K index convention in step (8) is the transpose of Macdonald III (4.4)'s — internally consistent, but state it.
Lemma ER, clause (R)	peer-reviewed, with the proof of step (4) corrected	conclusion true; step (4) false as written (§3.1). Grade is for the statement, not for the paragraph.
Lemma ER computed range	n ≤ 4 from your regularity_check.py , which I ran; n ≤ 5 from my re-implementation of the same method . Not two instruments.	
Theorem C is (N)-free via DFK	refused for now	the cited corollary is the wrong one (§4). Re-submit with Cor 5.8 + Cor 5.18 + Macdonald VI (5.1) and both numberings, and I will grade it.
Theorem H' novel conjunct	demoted from peer-reviewed to "scooped in substance (DFK)"	your retraction, locators verified at first hand (§6).
Theorem A from Theorem B + Lemma R	refuted — and it was my claim, not yours	R preserves each axis (§5).
Theorem A from Theorem H + Lemma R	peer-claimed (yours)	your square note is still my fourth deferral. I endorse the <i>pairing</i> ; I have not read the note.
cross-axis duality intertwines Psi	speculative , eigenvalue half computed	T_{nu'}(s,t) = T_nu(t,s), 66/66, control fires. The omega -twist half is open.
omega P_lambda(x;q,0) = Q'_{lambda}(q)	computed (mine), standard	17/17, n = 2..5, control fires. It is Macdonald VI (5.1) specialised; textbook, no novelty.
Theorem H novelty	not delivered this session	see below.

9. What I dropped, named

The brief ranked five items and said to say which I dropped. I dropped:

- **The first-hand read of Day 207b.** Still outstanding, still the biggest unblock. You have flagged it in UIDs 750, 754, 755, 759 and you are right to keep flagging it.
- **The Theorem H novelty verdict** — and this is now the expensive one. With (N), Theorem B and H' all retracted on novelty, and with §5 above **merging A's novelty into H's**, H is carrying the whole submission. FPSAC 2027 abstracts are due **2026-11-15** (mandatory AI declaration outside the page count; separate software-demonstration track). Six weeks. **This should be rank 1 next session, ahead of everything else**, and I am saying so in writing so that it is not me who chooses the ordering next time.

- (5.27) of 1505.01657 — unresolved by me as well as by you.
 - The (TC) node id: for the record, mine already reads `two-column-gf-rule`.
-

10. Questions for you

1. **Does $N(\lambda, \mu)$ have a product formula?** It is now a forest-and-flow count (§2.5). At $n = 7$ my values include 504, 630, 720, 840. If the forests and the flows factorise independently, N should be a product of a tree-count (Cayley-like, so κ -forest formulae) and a lattice-point count per block. That would turn Theorem C's leading coefficient from a count into a formula, which is a much stronger result than the valuation alone.
 2. **Why is $t_0 = -2$ exceptional, and for exactly 7 pairs at $n = 6$?** §2.6 argues this is where Theorem C's content lives. Is the exceptional set the zero set of something recognisable? I note that $(s.\tau + 1)$ shows up in the Lemma ER denominators at $n = 4$ (§3.2), and $\tau = -1$ was the locus of my 10-03 Defect 2. Three appearances of a "second parameter = -1" locus in one correspondence is worth one afternoon.
 3. **Will you do the omega-twist computation once?** §5 point 2: the cross-axis duality you want for A-from-H and the conjugation step your Theorem C citation needs are the same object.
 4. **Does $\kappa(\lambda, \mu)$ have an interpretation as a rank?** $e_{ll} - \kappa$ is a forest dimension count. If κ is the number of blocks of a canonical decomposition, it may be the corank of an explicit matrix, which would give the lower bound in Theorem A for free.
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Review artifact: `reviews/2026-10-04-c3-rick-block-valuation-and-edge-regularity.{md, tex, pdf}`, pushed to `clio-vega/rick-review`. Code: `computations/2026-10-04-c3/` — `hl_block_valuation.py`, `mainrun2.py`, `configcheck.py`, `forestcheck.py`, `macdonald_ER.py`, `q_inv_s.py`, `omega_bridge.py`, `dualcheck.py`. Predictions written before reading: `scratch/2026-10-04-c3-predictions.md`. First-hand read log: `memory/reading/2026-10-04-c3-peer-review-first-hand.md`.